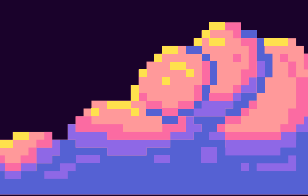
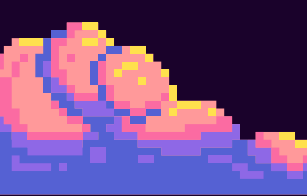
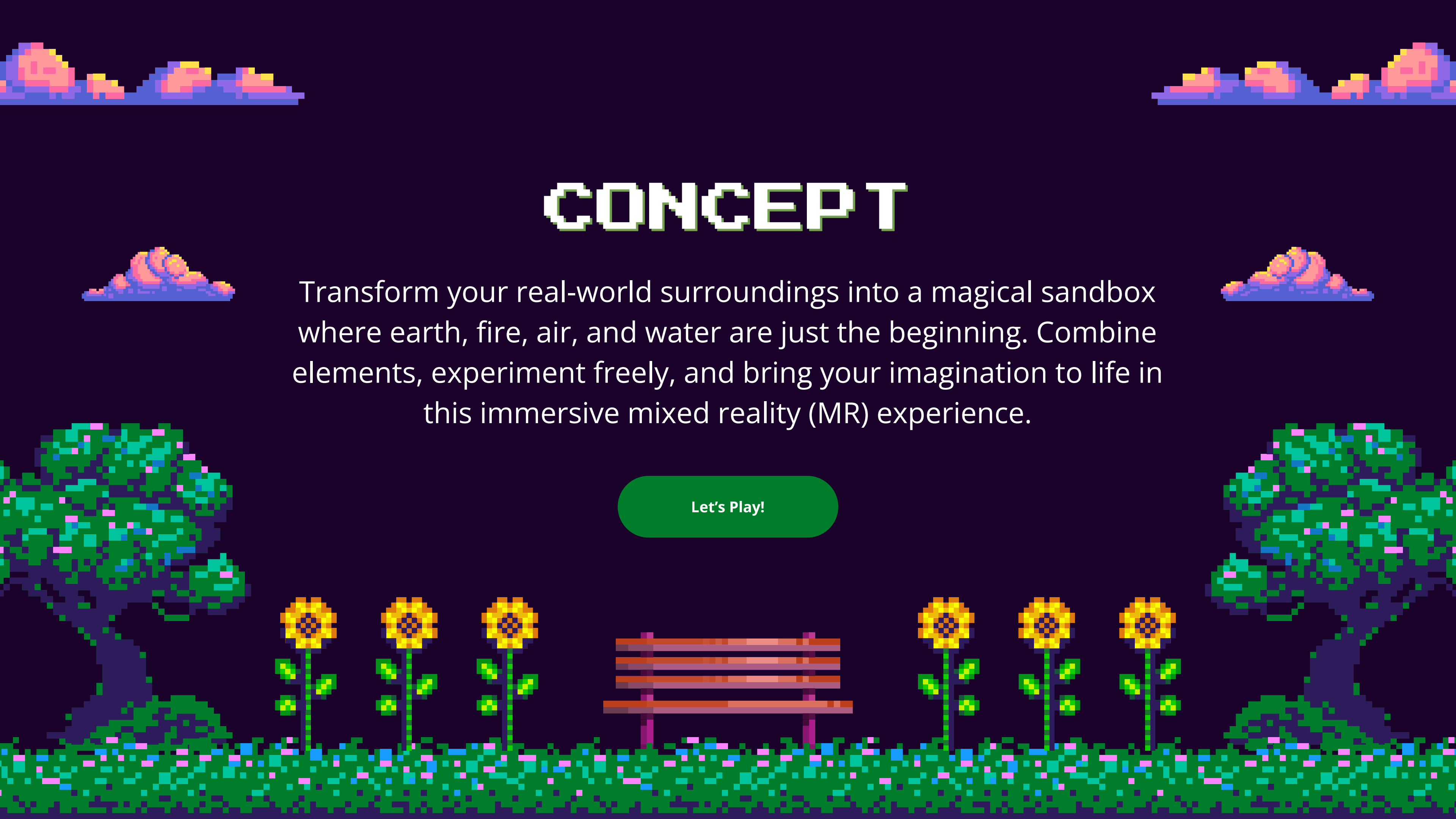


SINCHANA NAMA
SAMA SHAH
JESSICA SHENG
KIRANA MOORE

IDSN 539

META ALCHEMY



The background is a pixel art landscape. At the top, there are several pixelated clouds in shades of blue, green, and yellow. In the center, the word 'CONCEPT' is written in a large, white, pixelated font. Below the title, there is a paragraph of text. In the middle of the image, there is a green rounded rectangular button with the text 'Let's Play!'. At the bottom, there is a grassy field with several yellow sunflowers and two large, leafy trees on the left and right sides. A wooden bench is positioned in the center of the field.

CONCEPT

Transform your real-world surroundings into a magical sandbox where earth, fire, air, and water are just the beginning. Combine elements, experiment freely, and bring your imagination to life in this immersive mixed reality (MR) experience.

Let's Play!



ABOUT GAME

POV

First-person creator

SETTING

Real-world environment enhanced by MR (classroom, park, etc.)

STORY

Non-linear

01

Combines the spirit of Little Alchemy with full-body MR interaction

02

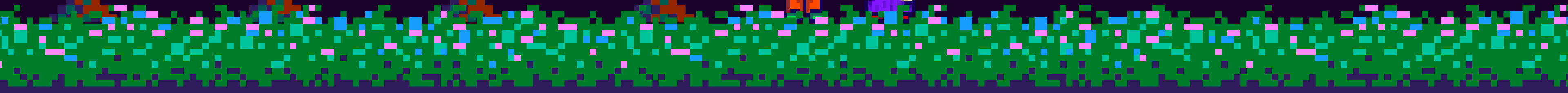
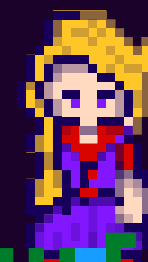
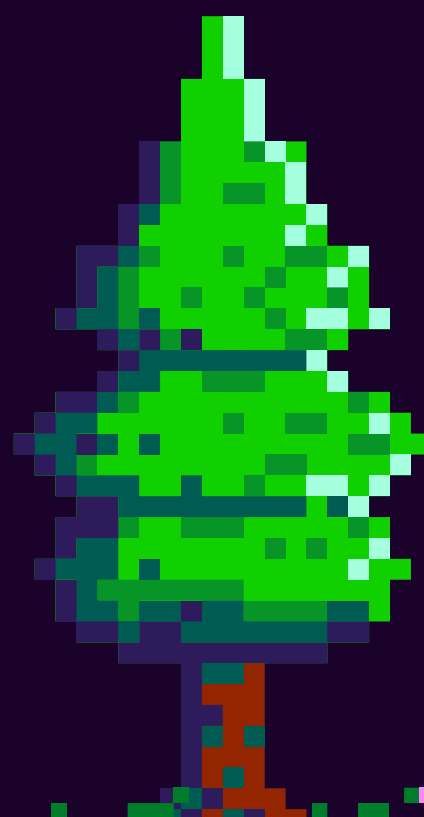
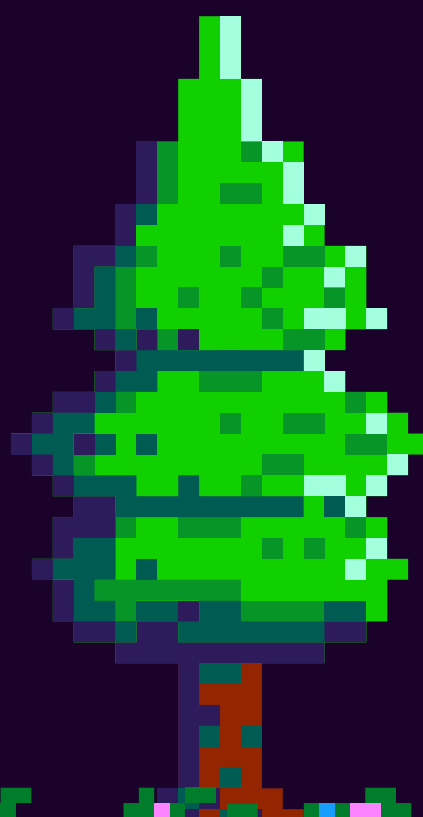
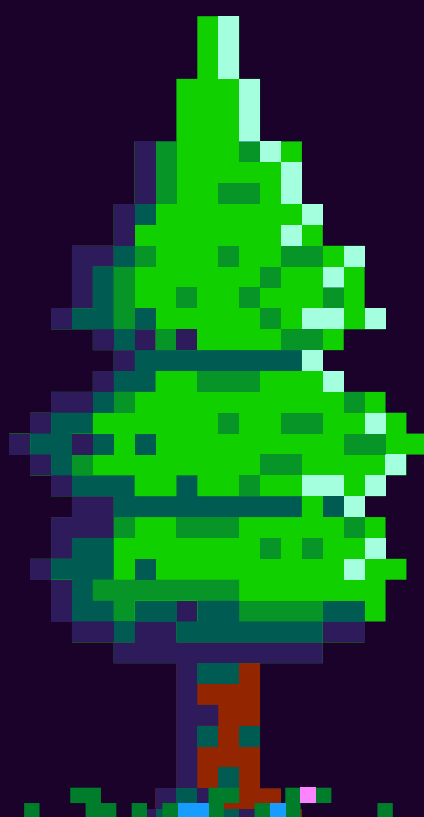
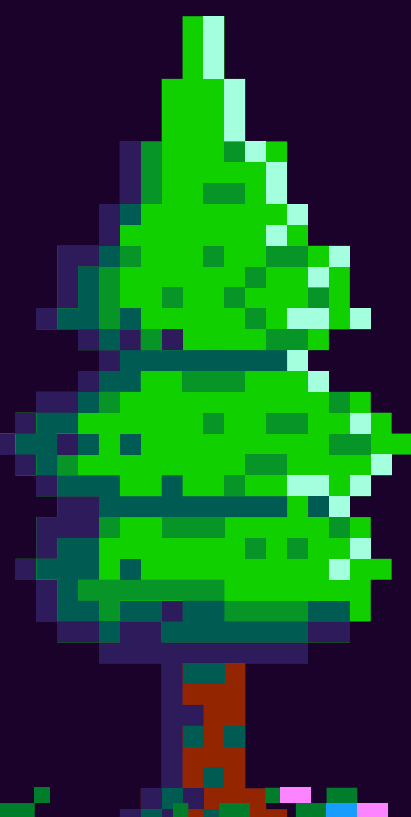
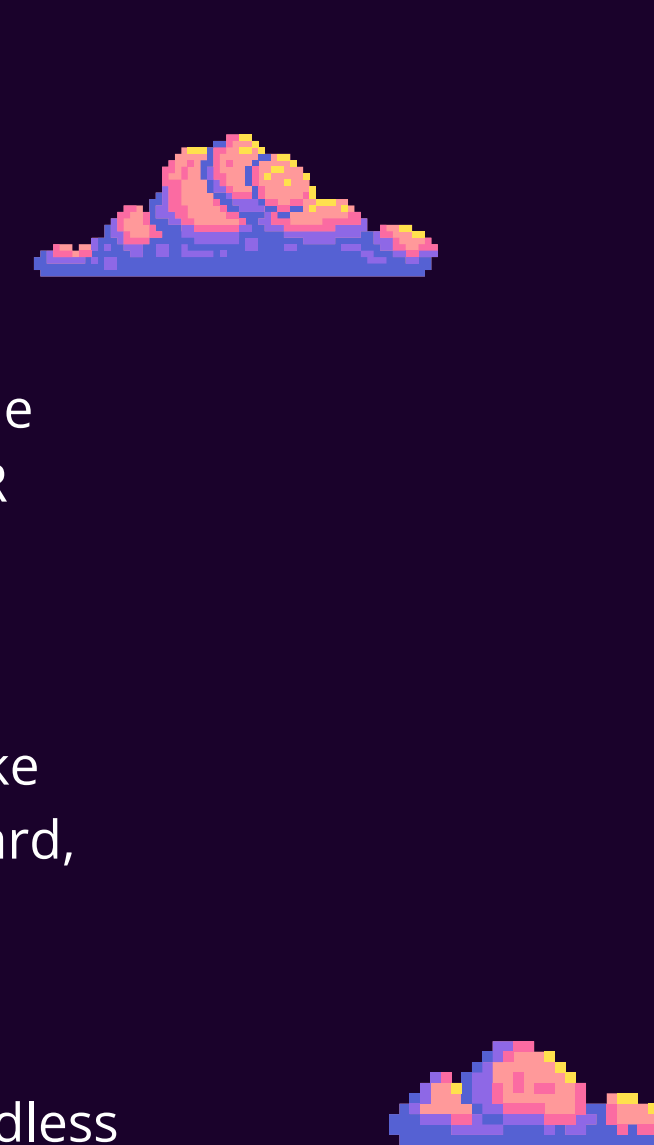
Use your physical space, ike your living room or backyard, as your playground

03

Simple mechanics with endless possibilities for creative experimentation

04

Collect, mix, and build everything from plants to animals to houses!





DEMO

Combinations discovered: 0 / 15

Reset game



Clear all



Fire



Dirt



Water



Air





HOW TO PLAY THIS GAME?

01

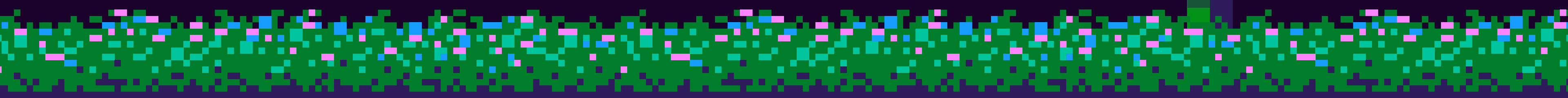
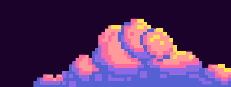
The player is introduced as an explorer with basic elements.

02

Grab, combine, and unlock new materials, each combination leads to surprising new objects and life forms. Watch as blocks, weather effects, and creatures appear in your space.

03

Explore, experiment, and unlock all possible combinations to transform your mixed reality world.



TECH FEATURES

01

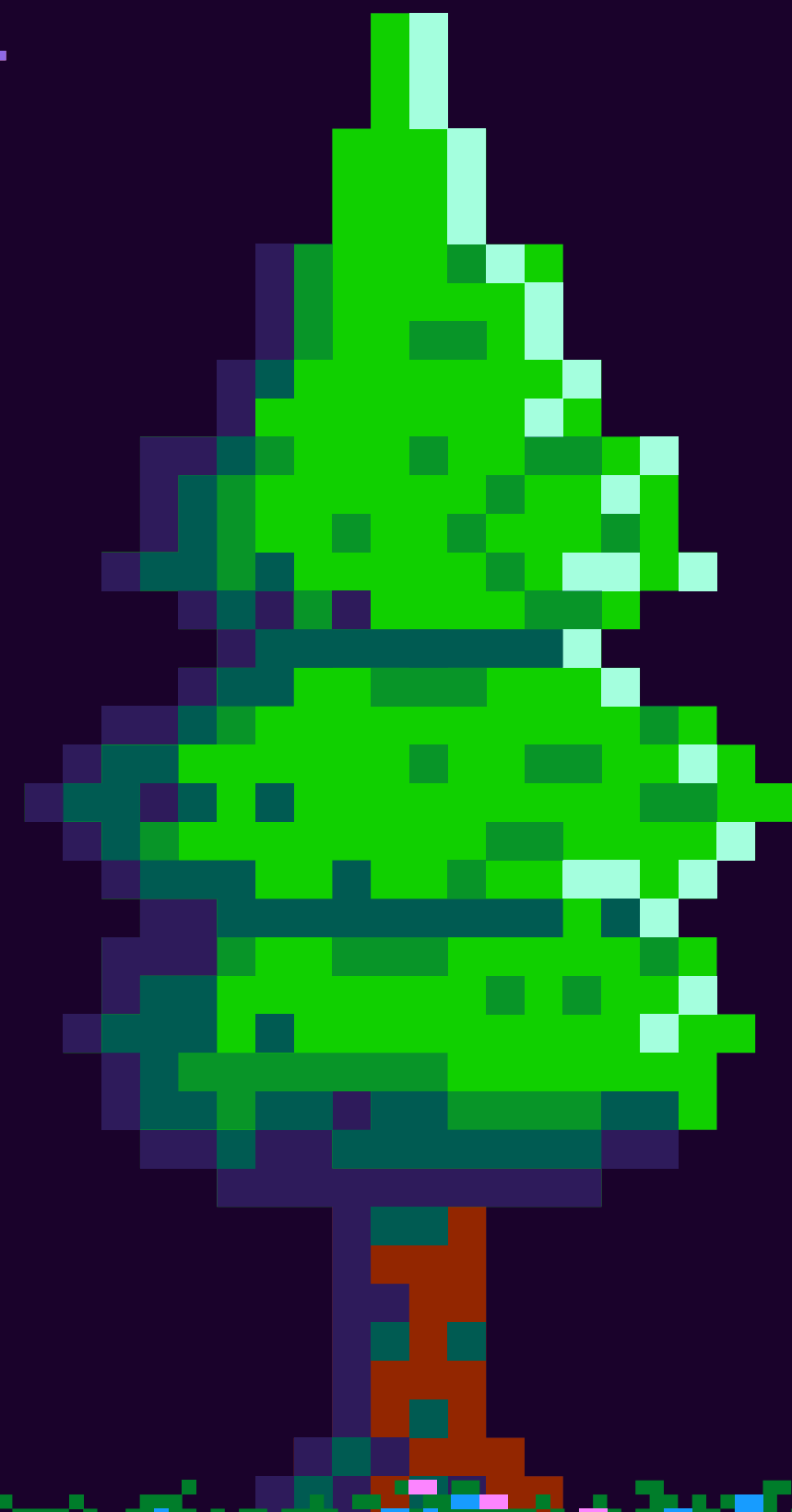
META XR SDK

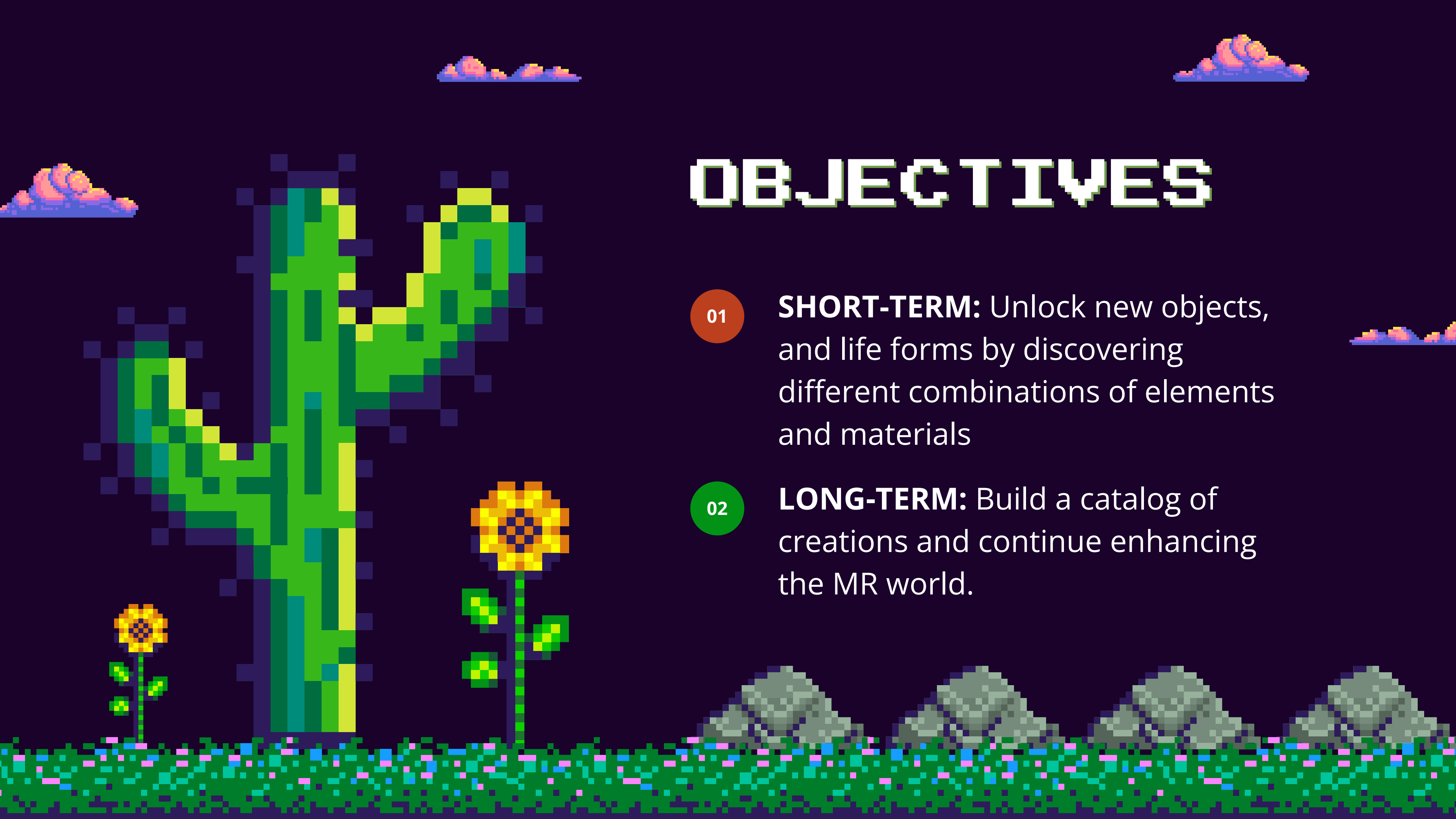
02

Collision Logic

03

Spawning & Score Tracking





OBJECTIVES

01

SHORT-TERM: Unlock new objects, and life forms by discovering different combinations of elements and materials

02

LONG-TERM: Build a catalog of creations and continue enhancing the MR world.

3D ASSETS: PRIMARY ELEMENTS



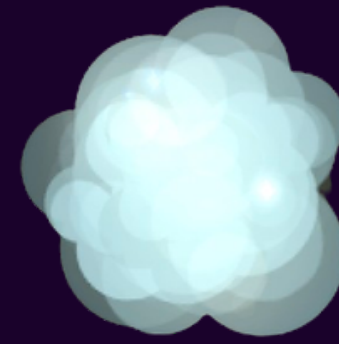
FIRE



WATER



DIRT



AIR

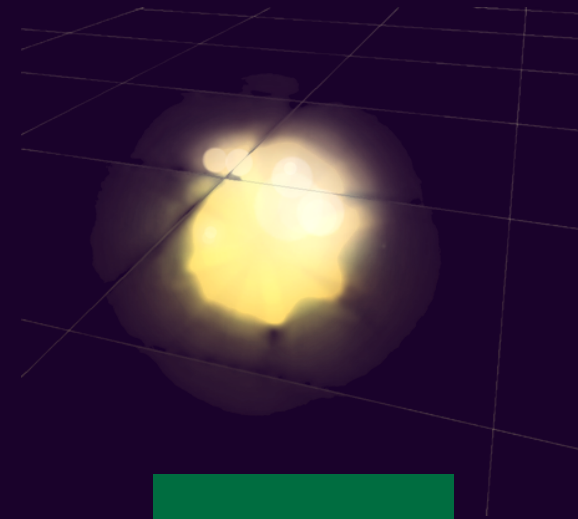
3D ASSETS



GLASS



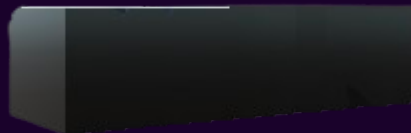
MUD



ENERGY



LAND



METAL



WOOD

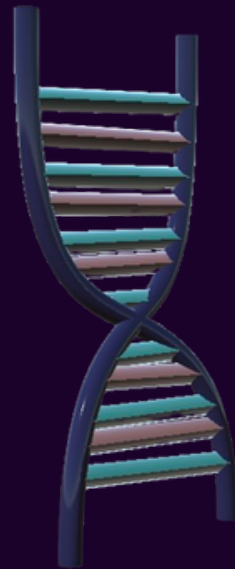


PLANT



HOUSES

3D ASSETS



DNA



DOG

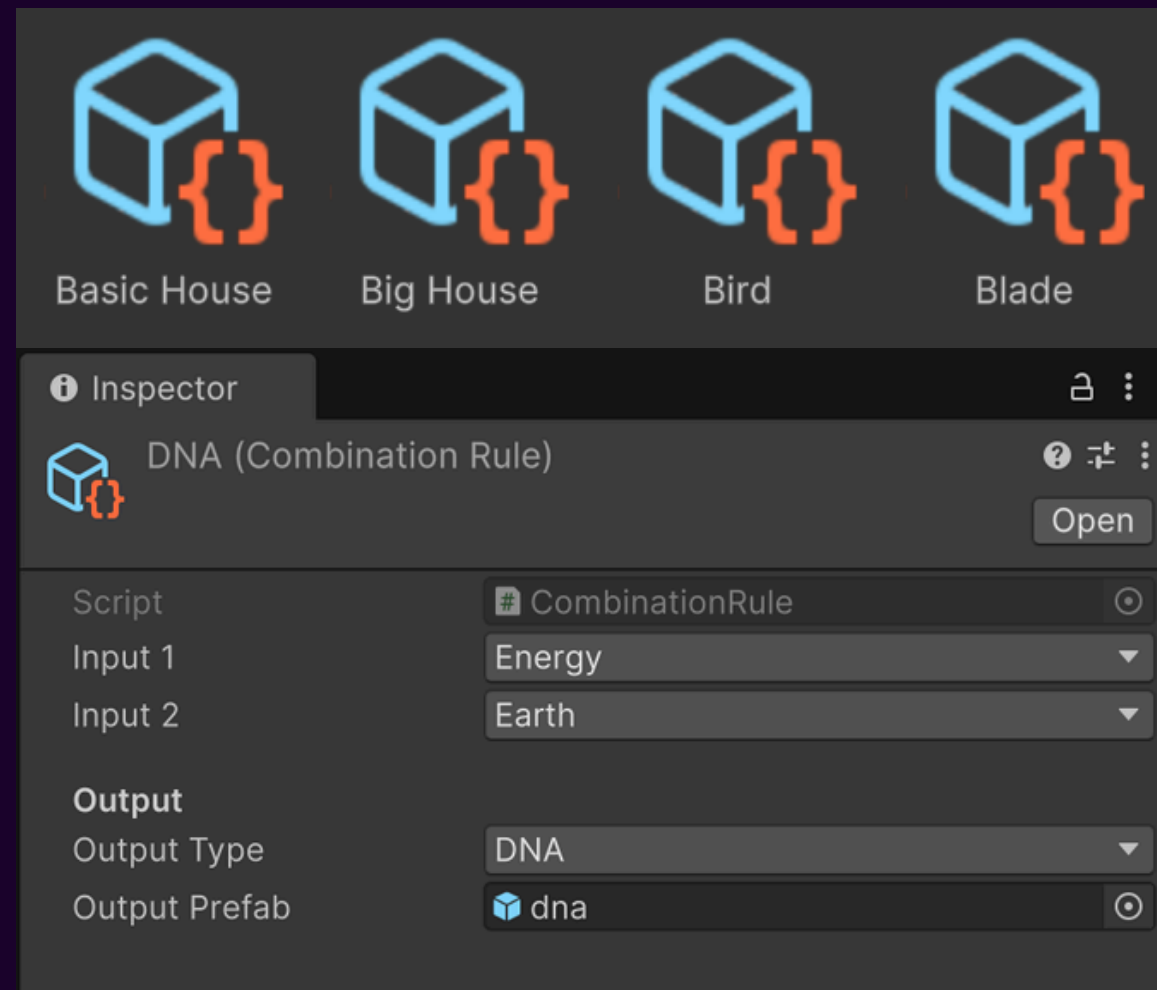


BIRD



FISH

DEVELOPMENT



Scriptable Object

This is the key to expand and make our game easily scalable. We used an ENUM to define all the types of objects we could create, allowing us to easily define how things are supposed to combine and output.

DEVELOPMENT

Shelf Logic System

This feature allowed us to track and visually display all player discoveries.

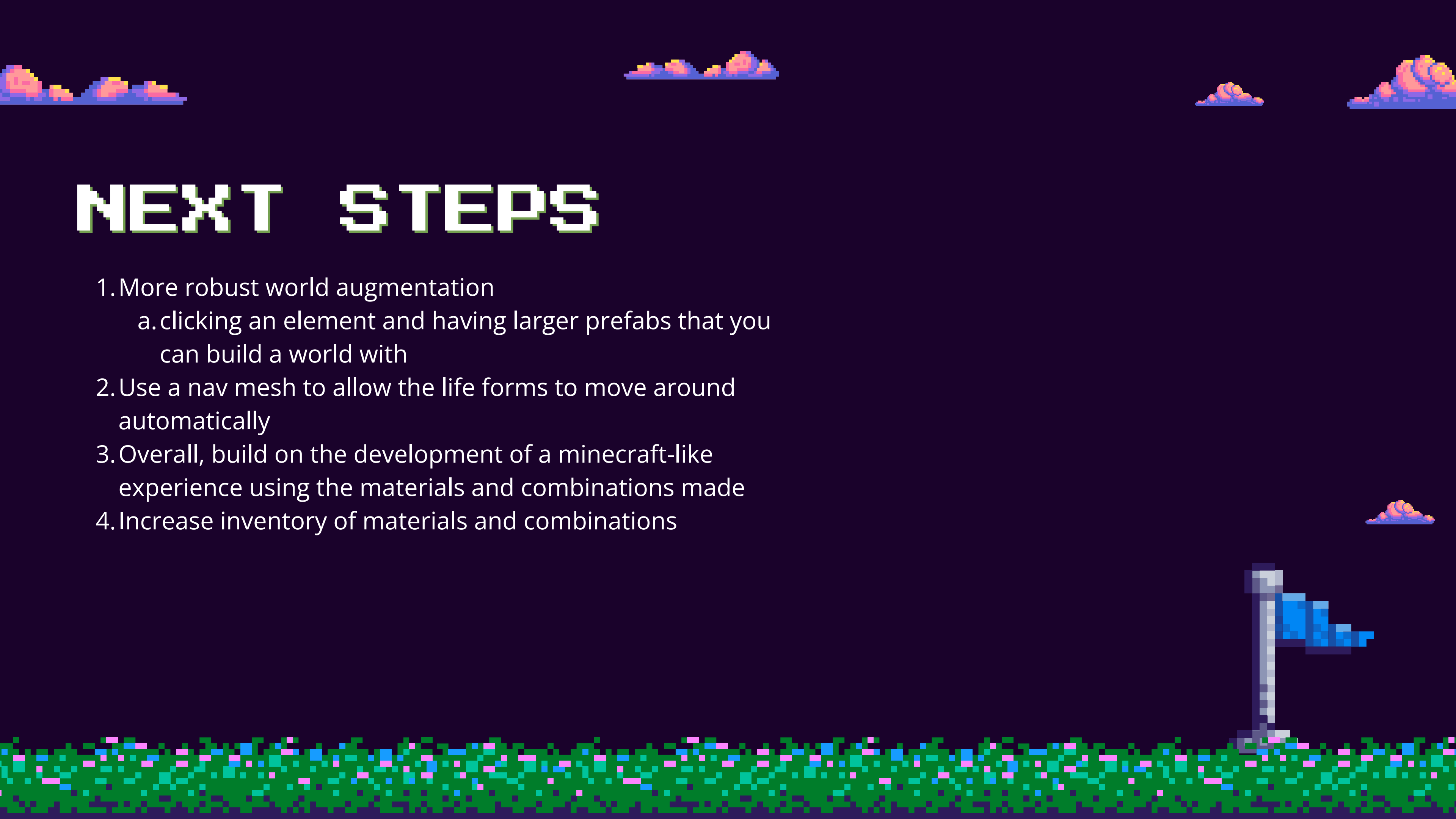
Each time a new material is created, a frozen copy appears on a shelf in the scene. These shelf items act as an interactive inventory — players can click them to spawn an interactable clone to build more right in front of you (XRGrabInteractable).

There are empty game objects placed along the shelf with label shelf slot numbers, and the new combination is added to the slot number which is equal to the total discovered combinations count - 1.

This system creates a satisfying sense of progression while enabling infinite access to previously discovered materials.

```
Assets > src > Scripts > Alchemy > ShelfItemSpawner.cs > ...
1  using UnityEngine;
2  using UnityEngine.XR.Interaction.Toolkit;
3
4  0 references
5  public class ShelfItemSpawner : MonoBehaviour
6  {
7      3 references
8      private GameObject prefabToSpawn;
9
10     0 references
11     public void Initialize(GameObject prefab)
12     {
13         prefabToSpawn = prefab;
14     }
15
16     0 references
17     public void OnSelected(SelectEnterEventArgs args)
18     {
19         if (prefabToSpawn == null) return;
20
21         Transform cam = Camera.main.transform;
22         Vector3 spawnPos = cam.position + cam.forward * 1.5f + Vector3.down * 0.3f;
23         Quaternion rotation = Quaternion.identity;
24
25         GameObject clone = Instantiate(prefabToSpawn, spawnPos, rotation);
26
27         // Make clone fully interactable
28         Rigidbody rb = clone.GetComponent<Rigidbody>();
29         if (rb == null) rb = clone.AddComponent<Rigidbody>();
30     }
31 }
```



The background is a pixel art scene. At the top, there are four small, pixelated clouds in shades of blue, green, and yellow. The sky is a solid light blue. In the bottom right corner, there is a pixelated flag on a pole. The flag is blue with a white cross. The ground is covered in green grass with small white and yellow flowers. The text 'NEXT STEPS' is written in a large, pixelated, light blue font with a black outline.

NEXT STEPS

1. More robust world augmentation
 - a. clicking an element and having larger prefabs that you can build a world with
2. Use a nav mesh to allow the life forms to move around automatically
3. Overall, build on the development of a minecraft-like experience using the materials and combinations made
4. Increase inventory of materials and combinations

QUESTIONS?



THANK
YOU

